# Fall 2025 EE582 Homework 05 Report

## Synchronous Machine Modeling and Transient Analysis

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# Question 1

**Problem Statement:** Develop a qd0 model (in per-unit) of a 325-MVA Hydro-Turbine Generator (book p. 220) in Simulink using basic blocks. The generator is to be connected to an ideal source (infinite bus) that supplies balanced three-phase sinusoidal voltages at rated 1 pu. Assume all variables are in pu, unless otherwise specified. Assume positive direction of $T_m$ is toward increasing the rotor speed $\omega_r$ and angle $\theta_r$. Choose an ode solver with the settings (error tolerances and step-size limits) that you think produces trustable results and is numerically efficient. Comment on your choice.

Table 1: Hydro Turbine Generator Parameters

| **Parameter** | **Value** |
|---|---|
| Rating | 325 MVA |
| Line-to-line voltage | 20 kV |
| Power factor | 0.85 |
| Poles | 64 |
| Speed | 112.5 r/min |
| Combined inertia of generator and turbine | $J = 35.1 \times 10^6$ J·s$^2$, or $WR^2 = 833.1 \times 10^6$ lbm·ft$^2$ |
| Inertia constant | $H = 7.5$ s |
| **Parameters in ohms and per unit:** | |
| $r_s$ | $0.00234\,\Omega$, 0.0019 pu |
| $X_{ls}$ | $0.1478\,\Omega$, 0.120 pu |
| $X_q$ | $0.5911\,\Omega$, 0.480 pu |
| $X_d$ | $1.0467\,\Omega$, 0.850 pu |
| $r'_{fd}$ | $0.00050\,\Omega$, 0.00041 pu |
| $X'_{fd}$ | $0.2523\,\Omega$, 0.2049 pu |
| $r'_{kq2}$ | $0.01675\,\Omega$, 0.0136 pu |
| $r'_{kd}$ | $0.01736\,\Omega$, 0.0141 pu |
| $X'_{lkq2}$ | $0.1267\,\Omega$, 0.1029 pu |
| $X'_{lkd}$ | $0.1970\,\Omega$, 0.160 pu |

**Solution:**

## Simulink Model Development

## ODE Solver Selection

# Question 2

**Problem Statement:** Assume zero initial conditions and $E_{xfd} = 0$. Implement a start-up transient similar to an induction machine (i.e., starting this synchronous machine from stall). Plot variables $i_{as}$, $T_e$, $\omega_r$. Comment on the results. Explain how it would be practical to bring a large machine like this online.

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**Solution:**

## Simulation Results

## Analysis of Start-up Transient

## Practical Start-up Procedure

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# Question 3

**Problem Statement:** Calculate the initial conditions that would correspond to an idle steady-state operation with $T_m = 0$ and excitation $E_{xfd} = 1.0$ pu, and set the initial conditions appropriately in your model. Implement the following transient study:

- At $t = 1$ s, the excitation $E_{xfd}$ is stepped down to 0.8 pu.
- Then, at $t = 7$ s, the excitation $E_{xfd}$ is stepped up to 1.2 pu.
- Then, at $t = 15$ s, the mechanical torque is stepped to $T_m = 0.85$ pu supplying power from the mechanical system to the electrical system, and the model is continued to run till $t = 25$ s.

Plot variables $i_{as}$, $i_{fd}$, $i_{kq}$, $T_e$, $\omega_r$, $\theta_r$ (in electrical degrees), as well as real power $P_e$ and reactive power $Q_e$. Based on the transients observed, briefly explain the effect of the excitation and torque on the active and reactive power and the type of transients seen.

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**Solution:**

## Initial Conditions Calculation

## Simulation Results

## Analysis of Excitation Effects

## Analysis of Torque Effects

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# Question 4

**Problem Statement:** Now assume that the q-axis damper winding is damaged. This can be implemented by setting the resistance of this winding to a very high value, e.g., set $r_{kq} = 1$ pu. Repeat what you did in part 3).

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**Solution:**

## Model Modification

## Simulation Results

## Comparison with Question 3

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# Question 5

**Problem Statement:** Now assume that the d-axis damper winding is damaged. Repeat what you did in part 3). No need to plot anything. Just make observations.

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**Solution:**

## Model Modification

## Observations

## Key Differences from Question 3

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# Question 6

**Problem Statement:** Based on the observations in parts 4) and 5), explain the effect and the role of each of these damper windings.

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**Solution:**

## Role of the q-axis Damper Winding

## Role of the d-axis Damper Winding

## Overall Function of Damper Windings

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